

ANAND BHALERAO

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Experience

FIS Global

Pune, India

Software Engineer 2

Feb 2026 - Present

- Designed and shipped **real-time session revocation** across multi-tenant systems serving **40+ banking clients** with zero downtime, containing a **\$2M fraud vector**; built atomic invalidation using Spring + PostgreSQL.
- Architected field-level AES encryption using a custom **MyBatis interceptor** for runtime SQL rewriting across 80% of application queries, **securing PII** across 300K+ records at **0.8ms overhead** paired with **blind indexing** to enable encrypted search with a **700x latency reduction**.
- Designed a **zero-downtime AES key rotation** framework with parallel, chunked migration and resumable execution, with encryption versioning to ensure **idempotent re-encryption** and transactional integrity across PII-bearing tables. Successfully migrated **500K+ encrypted records** per environment across 10+ environments.

Software Engineer 1

Jan 2025 - Jan 2026

- Led the **end-to-end Zelle SSO integration** for 12+ banking clients, implementing user-state **reconciliation** that synchronized external Zelle records with the bank's source of truth before SSO, alongside automated lifecycle workflows.
- Resolved **burst-traffic row-lock contention** by replacing a single-row sequence table with **block-based cached allocation**, eliminating request timeouts under load.
- Cut per-request payload from **90K rows to 10 rows** by pushing pagination, sorting, and filtering to the database layer, improving response times for 100K+ user datasets.

Associate Software Engineer

Jul 2024 - Dec 2024

- Eliminated recurring user lockouts by **redesigning multi-entity session handling**, enabling seamless account switching without redundant re-authentication.
- Integrated Outseer behavioral biometrics** into the authentication layer, enabling risk-based step-up authentication from client-side device signals.

Software Development Engineer Intern | **Certificate**

Jan 2024 - Mar 2024

- Developed an end-to-end Electronic File Delivery feature, building **8 Angular screens and 6 Spring REST APIs** to manage user preferences within a service-oriented architecture.
- Engineered a homepage interstitial for mandatory contact updates, implementing admin-controlled retry logic to ensure high compliance.

Projects

Distributed UPI Payment Simulator

Demo Video

Sarvatra Technologies

- Architected a **distributed UPI payment simulator** modeled on NPCI switch architecture using Spring Boot, Redis, and **RabbitMQ**.
- Implemented secure authentication using JWT for session management and **FIDO2/WebAuthn** for passwordless authentication with public-key cryptography.
- Built fault-tolerant payment processing with **idempotency**, **Saga-based distributed transactions**, and Redis-backed **Bloom filters** reducing database lookups by 92%.
- Instrumented end-to-end observability (OpenTelemetry + Prometheus) and load tested up to **1,800 TPS**, achieving **85ms p95 and 140ms p99** latency under sustained load.

Education

Pune Institute of Computer Technology

Jun 2020 - Jun 2024

B.E in Computer Engineering (CGPA - 8.94)

Pune, Maharashtra

Achievements

- 1100+ problems** solved across LeetCode (peak 1763), CodeChef (AIR 145, 496 in Starters), and Codeforces.
- Finalist in Smart India Hackathon '22.

Technical Skills

Languages: Java, Kotlin, Typescript

Application Development: Spring Boot, MyBatis, Hibernate, JPA, Angular, Compose Multiplatform

Databases & Cache: PostgreSQL, Oracle, MySQL, Redis

Infrastructure & Observability: RabbitMQ, OpenTelemetry, Prometheus